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Aircraft conceptual design computational tool for static aeroelastic analysis

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Abstract. *Static aeroelasticity examines the interaction between aerodynamic forces and structural deformation under steady-state conditions. This interaction can induce phenomena like divergence, control reversal, and load redistribution, all of which affect aircraft performance. In the conceptual design phase, accurately predicting load redistribution allows designers to optimize wing geometry for enhanced performance for highly flexible structures. To facilitate the wing design with a low-cost aeroelastic analysis, a 12-degree-of-freedom structural model was integrated with the Athena Vortex Lattice (AVL) method. The tool was used to analyze the NI-19 Wing of the SAE Aerodesign Nisus Team, from UFSC-CTJ, to evaluate the critical speed for divergence and load redistribution for leveled cruise flight. The results show that for leveled flight, the divergence occurs at a speed of* 73.25 *m/s, greater than the cruise speed, that is* 14.6 *m/s. The load redistribution for cruise flight reduced both the lift and the drag in 1.87% and 4.04%, respectively.*

Keywords: *Static Aeroelasticity, Vortex Lattice, Finite Element, Conceptual Design, Aircraft Strucutre.*

1. INTRODUCTION

The study of fluid-structure interaction has been of interest to engineers and researchers for years. In the field of aeronautical engineering, the science responsible for studying this interaction is aeroelasticity, which analyzes how aircraft stiffness and aerodynamic loads affect performance and safety. According to Hilger (2021), in recent years, the aeronautical industry has made increasing efforts to reduce the energy consumption and emissions of aircraft, creating a trend to develop lighter vehicles with a higher aspect ratio in order to reduce the drag they generate, and for these reasons, aircraft are becoming more flexible. Great examples of this are the unmanned HALE (High Altitude Long Endurance) aircraft that have been used for environmental monitoring, which have a very high aspect ratio, reaching 35 (Patil and Hodges, 2000) and high flexibility. In addition, the Boeing 787 Dreamliner, one of the company's most modern designs, is 50% made of composite material (Hale, 2006), and has an aspect ratio of 9.59 (AeroDatabase, 2024), making it one of the most flexible commercial aircraft today. It is notable, therefore, that aeroelastic phenomena have been increasingly influential in aeronautical design.

The reduction in structural rigidity in aircraft makes them more susceptible to static aeroelastic phenomena such as divergence and command reversal, which are extremely undesirable and should only occur outside the flight envelope of an aircraft (FAA, 2025) . With this in mind, in an effort to optimize aircraft structures while maintaining aeroelastic viability, studies have emerged on new structural wing designs, as presented by Qiao *et al.* (2019), inspired by bionic structures.

In recent years, the subject of aeroelasticity has been widely explored. Xie *et al.* (2012) presented an aeroelastic study of a very flexible HALE aircraft with a high aspect ratio, using a non-linear structural model due to the high deformations, and comparing the results obtained numerically with experimental results. Patil and Hodges (2000) also analyzed the aeroelastic behavior of very flexible aircraft, presenting a methodology for non-linear structural analysis.

Due to the structural nature of aeroelasticity, several optimization opportunities arise based on aeroelastic parameters. Bowman *et al.* (1989) carried out a structural optimization study of a wing based on command and lift efficiencies, resulting in a mass reduction compared to the initial model of 79%, and an increase in efficiencies of 80%. Also relevant, Dillinger *et al.* (2019) presented a study on the structural optimization of a composite wing based on aeroelastic criteria, focusing on the increase in stiffness and the stacking sequence of the laminates, comparing the influence of aerodynamic models of different levels of complexity on the results.

With the growing trend towards more flexible aircraft, it is becoming increasingly important to carry out aeroelastic analyses at an early design stage in order to avoid possible redesigns, increasing costs and delaying deadlines. In this

context, studies have emerged on structural and aeroelastic wing optimization while still in the preliminary design phase, also seeking computational savings, essential for early design phases, as demonstrated by Wan *et al.* (2014), who obtained a reduction of 89.1% in a wing's mass, while maintaining the structural and aeroelastic responses, considered constraints, at satisfactory levels. Going even deeper, Rajpal *et al.* (2019) applied a low computational cost optimization methodology, also in its preliminary phase, based on dynamic aeroelastic response, which is also capable of analyzing various load cases, updated according to the structural iteration, obtaining a mass reduction of 86.5% after just 42 iterations.

By their very nature, aeroelastic problems require well-defined aerodynamic and structural models to be solved; iteratively, the loads are obtained, followed by the deformations in the structure, which are then inserted into the aerodynamic model, recalculating the loads, and consequently, new deformations occur, the cycle only ends when the equilibrium condition is reached. Computationally, there are two main approaches for solving the problem: strongly coupled, which is more precise and costly, when both the aerodynamic and structural models are integrated directly, and weakly coupled (decoupled), where the models are solved in parallel, and the results are only passed on as constantly updated parameters. Scholten and Hartl (2021) present a methodology for a weakly coupled aeroelastic model, comparing the performance of the model with a strongly coupled approach, the decoupled model showed a considerably lower computational cost, for a result similar to the coupled approach.The present work uses a weakly coupled approach, aiming at computational performance, allowing for quick, and relatively precise analysis during a conceptual, or preliminary project phase.

2. METHODOLOGY

In this section, the methodology and its implementation will be described. The coupled iterative method proposed in this investigation consists of a Vortex Method (*VLM*) for potential flow as aerodynamic model, presented in section 2.2 and the 12 Degrees of Freedom (*DOF*) Finite Element as structural model presented in section 2.3 The coupling was performed using a weakly coupled iterative method, the implementation is described in section 2.4 The geometry investigated is described at section 2.1

2.1 Wing Geometry

The wing model used in this study is a rectangular wing developed by Nisus Aerodesign Team for the SAE Aerodesign Brazil 2019 competition. The wing geometrical parameters are listed in the Table 2.1 Schenckel (2020) presented a structural analysis and sizing of this wing using Finite Element Analysis in *Ansys*, the results will be used as reference in this study. The wing geometry is shown below.

Airfoil	E423
Wingspan (B)	2.2 m
Chord (c)	0.49 m
Incidence (α_G)	0º

Table 1. Wing Geometry

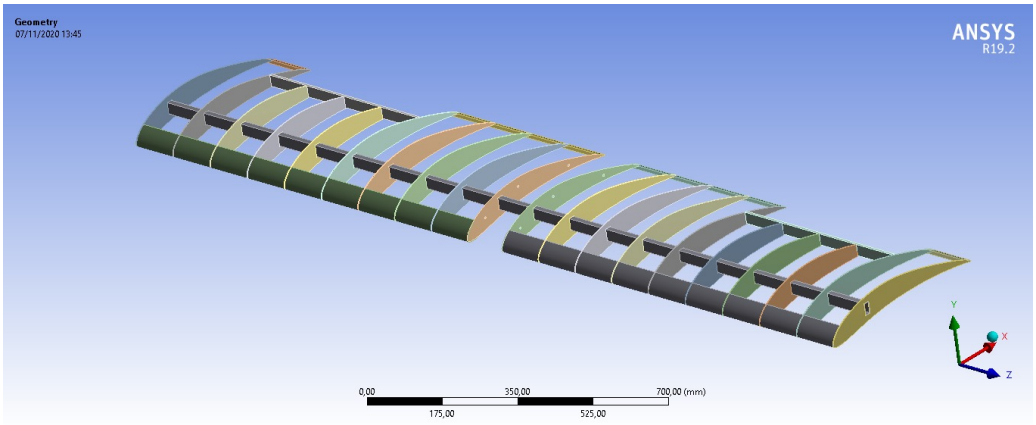


Figure 1. Wing Geometry (Schenckel, 2020)

2.2 Aerodynamic Model

To perform the aerodynamic analysis, and obtain the flight loads, the *AVL* (*Athena Vortex Lattice)* software was used. The software is an aerodynamic solver based on potential flow, developed by Drela (2014), with an implementation of the

Vortex Lattice Method. Given the boundary condition of non-penetrability of the surfaces:

$$\nabla(\phi + \phi_\infty).\hat{\eta} = 0 \tag{1}$$

where ϕ is the perturbation potential and ϕ_∞ is the free stream potential and $\hat{\eta}$ is the surface normal vector, this panel method distributes on the wing several elementary solutions along the wing mean surface, and its strengths are calculated defining the linear system given by:

$$[AIC]\{\Gamma\} = -\left\{\vec{U_0}.\hat{\eta}\right\} \tag{2}$$

where $[AIC]$ is the aerodynamic influence coefficient matrix, $\vec{U_0}$ is the free stream velocity vector and $\{\Gamma\}$ is the vector of elementary solutions strength. In that formulation, the normal vector and the elements of $[AIC]$ matrix are changed by the wing deformation.

To allow the integration of the structural and aerodynamic models, the wing geometry is defined by 12 sections where the elastic torsion is considered. As result, the AVL software integrates the lift, induced drag and pitch moment coefficients (c_l , c_d and c_m) for each strip along the wing span. The strips are defined by a chordwise element distribution at a constant span location. So, considering a defined dynamic pressure, the lift, drag and torsion distributions are calculated and used as loads in the structural finite element model described at section 2.3

2.3 Structural Model

To obtain the deformations of the wing model, a 12 *DOF* Finite Element Model (*FEM*) based on Beam Elements, also known as a Frame Element (Liu and Quek, 2003) was implemented in *Python* , allowing to obtain spacial displacements and rotations. The figure 2.3shows the FEM element, where the coordinate system is on aircraft reference frame.

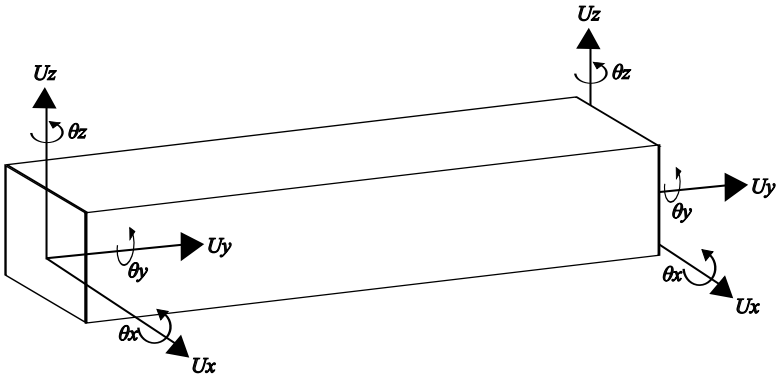


Figure 2. Frame Element

The finite element model, defined by the linear system shown in equation 3, has its stiffness matrix constructed by the collocation method for a linear mesh of elements over the elastic line of the wing. In that equation, the $[K]$ is the stiffness matrix, $\{U\}$ is the *DOF's* displacements and $\{F(\Gamma)\}$ is the nodal equivalent loads vector.

$$[K]\{U\} = \{F(\Gamma)\} \tag{3}$$

It can be seen that the loads are dependent of the aerodynamic results, represented by Γ. To construct that load vector, the FEM loads were considered constant load distributions over each element, as shown in equations 4, 5 and 6,

$$L = \begin{Bmatrix} F_{z1} \\ M_{x1} \\ F_{z2} \\ M_{x2} \end{Bmatrix} = \begin{Bmatrix} \frac{L' \times l_{elem}}{2} \\ \frac{-L' \times l_{elem}^2}{12} \\ \frac{L' \times l_{elem}}{2} \\ \frac{L' \times l_{elem}^2}{12} \end{Bmatrix} \tag{4}$$

where L is the lift bending loads, $F_{z[1,2]}$ are the equivalent shear loads at element nodes, $M_{x[1,2]}$ are the equivalent bending moments at element nodes, L' is the constant lift distribution on the element and l_{elem} is the element length.

$$D = \begin{Bmatrix} F_{x1} \\ M_{z1} \\ F_{x2} \\ M_{z2} \end{Bmatrix} = \begin{Bmatrix} \frac{D' \times l_{elem}}{2} \\ \frac{-D' \times l_{elem}^2}{12} \\ \frac{D' \times l_{elem}}{2} \\ \frac{D' \times l_{elem}^2}{12} \end{Bmatrix} \tag{5}$$

where D is the drag bending loads, $F_{x[1,2]}$ are the equivalent shear loads at element nodes, $M_{z[1,2]}$ are the equivalent bending moments at element nodes and D' is the constant drag distribution on the element.

$$T = \begin{Bmatrix} M_{y1} \\ M_{y2} \end{Bmatrix} = \begin{Bmatrix} \frac{T' \times l_{elem}}{2} \\ \frac{T' \times l_{elem}}{2} \end{Bmatrix} \tag{6}$$

where T is the pitch moment torsion loads, $M_{y[1,2]}$ are the equivalent pitch torsion at element nodes and T' is the constant pitch torsion distribution on the element, defined by equation 7.

In this study, the structural behavior of the wing was modelled after the geometric and mechanical properties of the wingspar, in order to maintain a stiffness equivalent to the original model. The wing's elastic axis was considered to be in 30% of the root chord along the wingspan. The structural properties of the model are presented below.

Material	Carbon Fiber-Epoxy
Young Modulus (E) [MPa]	59200
Shear Modulus (G) [MPa]	17500
Polar moment of inertia (J) [m^4]	1.52175e-08
Moment of inertia I_x [m^4]	1.12675e-08
Moment of inertia I_z [m^4]	3.950e-09
Area (A) [m²]	9.0e-05
Elastic Axis [% Chord]	0.3

Table 2. Structural Properties

2.4 Load Redistribution

To perform the aeroelastic analysis, the two solvers (the aerodynamic and the structural) were integrated by an iterative strategy. First, the strip forces are determined in *AVL* , which are used to obtain the lift, and torsional moment distributions along the wing, using the typical aeroelastic section model (Coutinho, 2023), shown in figure 3.

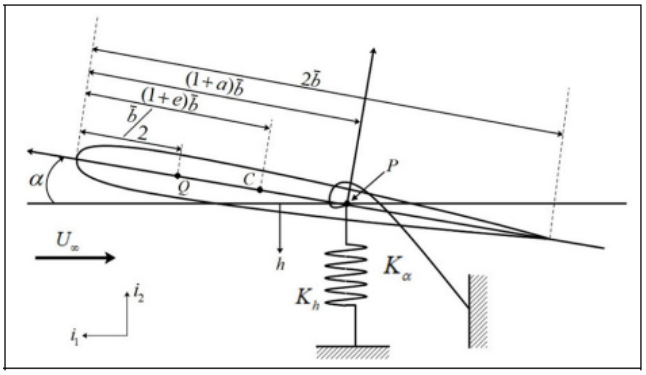


Figure 3. Typical aeroelastic section

$$T(y) = q.c(y)(cl(y).ec(y) + cm(y).c(y)) \tag{7}$$

where q is the dynamic pressure, $c(y)$ is the section chord, $cl(y)$ is the section lift coefficient, $ec(y)$ is the section lift eccentricity and $cm(y)$ is the section pitch moment coefficient. In equation 7, the aerodynamic coefficients are influenced by the structure deformation.

The aeroelastic model is defined, in a simple way, as a system of equations 2, 3 and 7. That system is solved by an iterative process illustrated in figure 4. The initial condition considers no deformation of the wing, then the aerodynamic geometry is defined and solved using AVL software, resulting in the aerodynamic loads. After that, the equation 7 is used to define the torsional load and the loads are applied in the FEM model, resulting in the nodal displacements. If the nodal displacements changes are greater than a tolerance ($1e-2$ for torsion and $1e-3$ for bending displacement), the process is recalculated with the geometrical torsion updated.

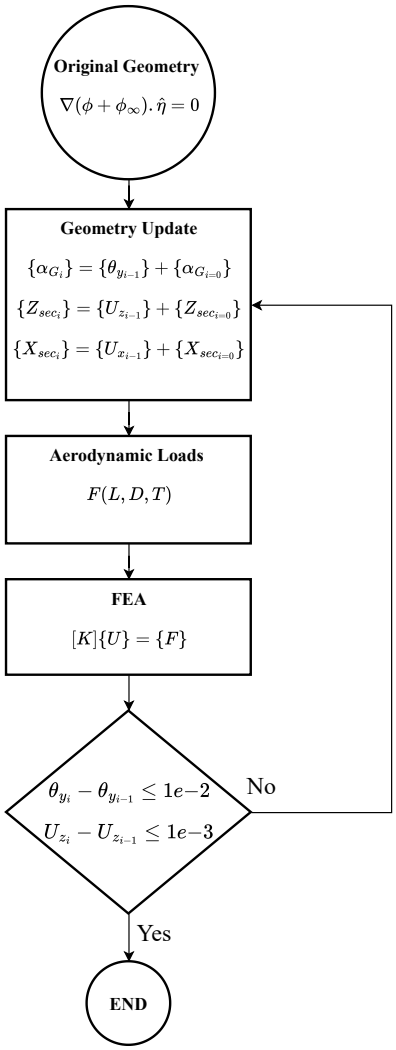


Figure 4. Aeroelastic analysis algorithm

2.5 Wing Aeroelastic Divergence

When faced with a certain value of dynamic pressure, the wing may fail structurally due to excessive torsion, a phenomenon known as aeroelastic divergence. The wing tip twist in a rectangular cantilevered wing can be expressed as (Wright and Cooper, 2008):

$$\theta_T = \frac{3qec^2s^2a_w}{6GJ - 2qec^2s^2a_w}\theta_0 \tag{8}$$

being s the semispan of the wing, a_w its lift-curve slope, and θ_0 the root incidence. The wing is considered to diverge when the twist reaches infinity, in this condition, the divergence pressure is found as

$$q = \frac{3GJ}{ec^2s^2a_w} \tag{9}$$

To analyze this, the wing tip twist was calculated through load redistribution, explained in section 2.4, for increasing velocities. Despite, theoretically, divergence occurring when the twist reaches infinity, failure will be considered to occur sooner, as large twists are unacceptable, 40° were considered the failure criteria.

3. RESULTS

Two different cases will be investigated for the geometry presented: the wing divergence and load redistribution in cruise conditions. The load redistribution case is presented in section 3.2and the divergence case in section 3.1

3.1 Divergence

The divergence analysis is shown in figure 5. It can be seen that the method predicted a divergence speed of $73.25m/s$ ($3286.63Pa$ of dynamic pressure) and the theoretical prediction was $14553.35Pa$. From the previous study (Coutinho, 2023), the theoretical was lower than the numerical one, and in this case it was the opposite. The reason for this is not yet understandable, however, it is believed that the analytical solution used could fail to correctly represent the behavior of a low speed, high lift airfoil, such as the one used in this study. Nevertheless, the two predictions are significantly higher than the cruise speed, so it is considered safe to operate that wing at the defined cruise speed.

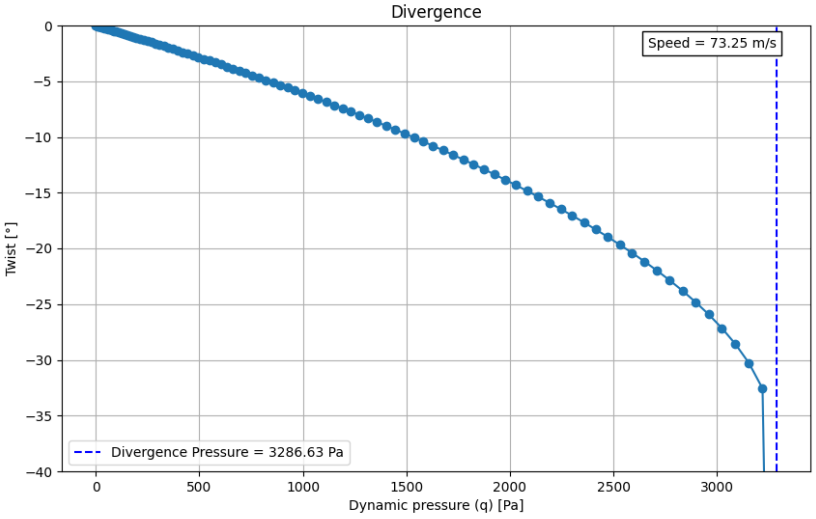


Figure 5. Divergence Analysis

3.2 Load Redistribution

As a load case, the wing was analyzed in $\alpha = 10^o$, in a freestream velocity of 14.6m/s, and in an air density of 1.225kg/m³. The load redistribution analysis converged in 3 iterations, obtaining the elastic lift and pitch moment distributions, as well as the wing deflection, and twist, all being shown below. Figure 6 shows the bend deflection results and the figure 7 shows the torsion distribution. The figure 8 shows the results for the lift distribution of the wing without deformation, and after aeroelastic analysis, and the torsional loads components.

From figure 7, it can be seen that the wing section incidence reduces to the wing tip and its slope reduces close to the tip. The same behavior was obtained on Coutinho (2023) investigation, but in that case, positive values. That reduction in wing incidence reduced the lift values, as seen in figure 8. As a result, the total Lift of the wing was reduced by the torsion, been $CL = 1.4637$ for the undeformed wing and $CL = 1.43632$ for the deformed one, a 1.87% reduction. In the

same way, the induced drag changed from $CD = 0.15112$ for the undeformed wing, to $CD = 0.14502$ in the deformed one, a 4.04% reduction. That changed the aerodynamic efficiency from $L/D = 9.6856$ to $L/D = 9.9043$, increasing 2.26%. Despite losing lift, it can be seen that the reduction in induced drag caused a significant increase in aerodynamic efficiency.

It can be seen that the measured deflection was 21.59mm, a value lower than the predicted by the Schenckel (2020) investigation, which was 31.2mm. This could be caused by the reduction in bending moment related to the lift reduction by the wing torsion. The aeroelastic analysis evidenced that including the change in loading due to the fluid-structure interaction can be very advantageous, providing a more accurate analysis, and possibly conceiving lighter structures.

In addition, it is important to remark that the change in the pitch, and torsional moment were not much significant in comparison to lift and drag, both went from $CM = -0.6194$ and $T = -6.43$Nm in the undeformed wing, to $CM = -0.6146$ and $T = -6.51$Nm in the deformed one, generating a -0.719^o twist in the wing tip.

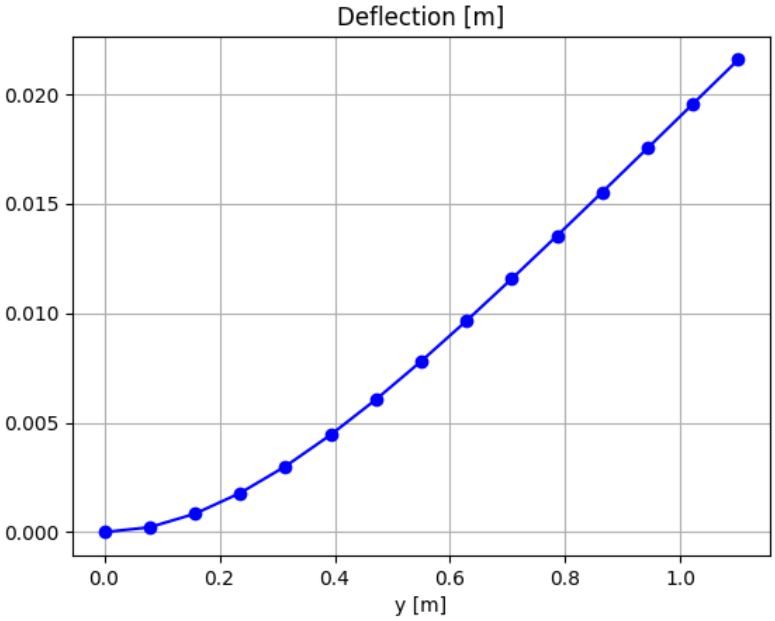


Figure 6. Deflection results

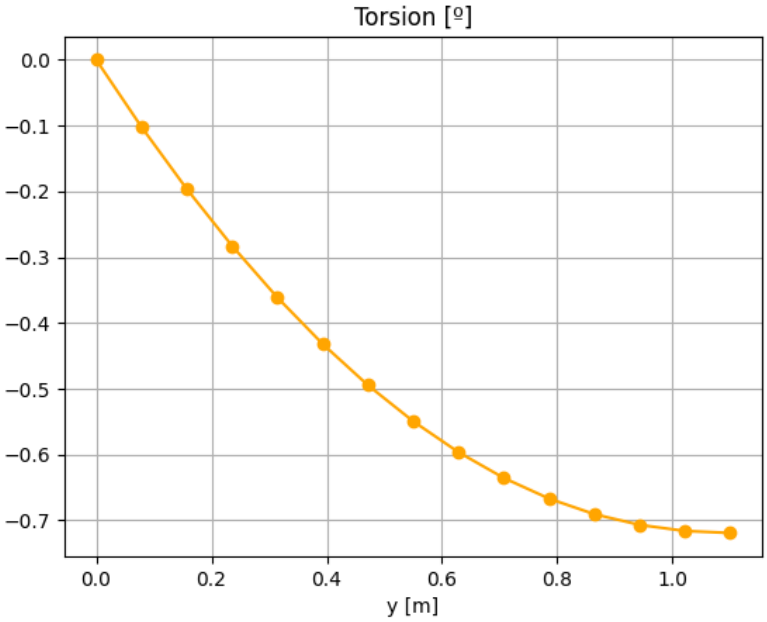


Figure 7. Torsion results

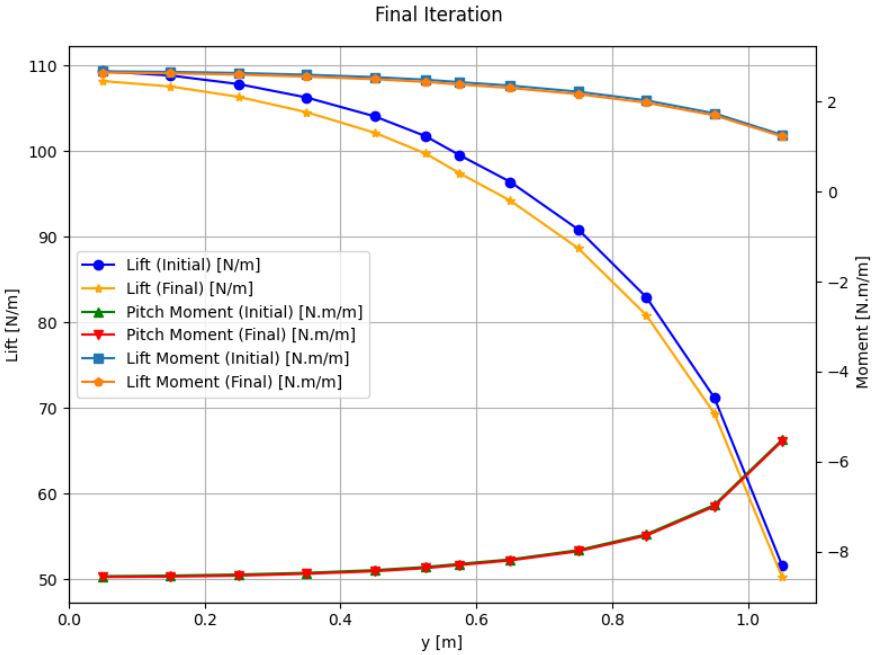


Figure 8. Lift and Moment distribution in final iteration

4. CONCLUSION

In this investigation, a static-aeroelastic method was defined using a 12 DOF beam for bending-torsion model, integrating it to the AVL (Athena Vortex Lattice) aerodynamic model. The iteration is closed by the typical section considering the pitch moment and the lift's torsional torque over the wing span. The wing divergence and the load redistributions were evaluated for the NI-19 wing. The results show that the wing divergence predicted by the method occurs for a speed of 73.25m/s, a value greater than the cruise speed. The load redistribution predicted that the wing tip displacement is about 21.59mm for bending, and the torsion was -0.719^o. The total lift was 1.87% lower compared with the rigid case, and The induced drag was 4.04% lower, in comparison with the rigid case, generating a 2.26% increase in aerodynamic efficiency.

5. ACKNOWLEDGEMENTS

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